Order summary: 2 lamps (generated by `python3 bom/build_bom.py --sets 2`; do not edit by hand)

Indicative prices from Farnell UK listings and seller pages, 2026-11. **Matt places every order.** After ordering, record the actual codes and prices in `bom/ORDER_ACTUALS.md` (handoff 02).

1. Farnell UK basket

Upload `bom/farnell_quickpaste.txt` (Quick Paste / BOM upload). 31 lines, quantities include spares and MOQ rounding. Lines without an order code are listed by MPN: pick that exact MPN on the site (substitution rules in handoff 02).

Farnell code	MPN	Manufacturer	Qty	Needed	Line GBP ex VAT	What
3648642	USB4125-GF-A	GCT	3	2	1.77	USB-C receptacle
(search MPN)	MF-MSMF110/16X-2	Bourns	5	2	2.00	Resettable fuse 1.1 A hold / 2.2 A trip
(search MPN)	MCWR06X5101FTL	Multicomp Pro	50	4	0.50	Resistor 5.1 kOhm 1% 0603
2691565	SMF5.0A	Littelfuse	5	2	0.75	TVS unidirectional 5.0 V standoff
(search MPN)	CL21A106KOQNNNE	Samsung	20	8	1.60	MLCC 10 uF 16 V X5R 0805 (0603 10 V part keeps only 2.9 uF at 5 V)
1332158	MCP73831T-2ACI/OT	Microchip	3	2	1.77	Li-ion/LiPo linear charger 4.20 V
(search MPN)	MCWR06X3901FTL	Multicomp Pro	50	2	0.50	Resistor 3.9 kOhm 1% 0603 (charge current 256 mA)
(search MPN)	MCWR06X1001FTL	Multicomp Pro	50	4	0.50	Resistor 1 kOhm 1% 0603
(search MPN)	APT1608SURCK	Kingbright	4	2	0.60	Hyper-red chip LED 0603
9492615	S2B-PH-SM4-TB(LF)(SN)	JST	10	2	4.80	JST PH 2-pin side-entry SMD header (battery)
2543533	DMG2305UX-7	Diodes Inc	10	8	3.00	P-MOSFET -20 V -4.2 A
2447230	MCWR06X1002FTL	Multicomp Pro	50	8	0.50	Resistor 10 kOhm 1% 0603
(search MPN)	S1J-E3/61T	Vishay	5	2	0.60	Silicon rectifier 1 A 600 V
(search MPN)	MCWR06X1004FTL	Multicomp Pro	50	2	0.50	Resistor 1 MOhm 1% 0603
1759122	MC0603B104K500CT	Multicomp Pro	50	6	0.50	MLCC 100 nF 50 V X7R 0603
9575146	PCM12SMTR	C&K	3	2	2.55	Mini slide switch SPDT
3018703	ATTINY3216-SN	Microchip	3	2	3.12	8-bit AVR MCU
3013385	CL10A105KB8NNNC	Samsung	10	2	0.30	MLCC 1 uF 50 V X5R 0603
3941596	PTS645SM43SMTR92 LFS	C&K	10	6	1.90	Tactile switch 6x6x4.3 mm SMD

Farnell code	MPN	Manufacturer	Qty	Needed	Line GBP ex VAT	What
(search MPN)	MCWR06X1003FTL	Multicomp Pro	50	2	0.50	Resistor 100 kOhm 1% 0603
(search MPN)	MCWR06X3301FTL	Multicomp Pro	50	4	0.50	Resistor 3.3 kOhm 1% 0603 (I2C pull-up)
1830839	SM04B-SRSS-TB(LF)(SN)	JST	3	2	1.35	JST SH 4-pin side-entry SMD header (Qwiic pinout
3498849	BXEN-27S-11M-3C-00-0-3	Bridgelux	10	8	1.00	Mid-power white LED 2835
3498853	BXEN-50S-11M-3C-00-0-3	Bridgelux	10	8	1.80	Mid-power white LED 2835
2322115	150141M173100	Wurth Elektronik	3	2	1.32	RGB LED PLCC-4 3528
(search MPN)	RC0402FR-072K94L	Yageo	50	2	0.50	Resistor 2.94 kOhm 1% 0402 (LP5012 RIREF: 105 x 0.7 V / 2.94 k = 25.0 mA/channel)
(search MPN)	GRM155R61E105KA12D	Murata	10	4	0.30	MLCC 1 uF 25 V X5R 0402 (LP5012 VCC / VCAP)
(search MPN)	GRM188R61A106KE69D	Murata	10	2	0.50	MLCC 10 uF 10 V X5R 0603 (LED board bulk)
1439485	MCP9800A0T-M/OT	Microchip	5	2	2.75	I2C temperature sensor
(search MPN)	RC0402FR-072K4L	Yageo	50	2	0.50	Resistor 2.4 kOhm 1% 0402 (LP5012 EN pull-up)
(search MPN)	GRM155R71C104KA88D	Murata	50	2	0.50	MLCC 100 nF 16 V X7R 0402

Farnell subtotal ≈ £39.28 ex VAT (≈ £47.14 inc VAT).

1b. Electronic parts Farnell does not stock

Order these from the supplier named (`bom/digikey_order.csv`): DigiKey UK ships cut tape from quantity 1.

Supplier	Code / search	MPN	Manufacturer	Qty	Needed	Line GBP ex VAT	What
DigiKey UK	LP5012RUKR (cut tape) - digikey.co.uk	LP5012RUKR	Texas Instruments	3	2	2.70	12-ch I2C constant-current RGB LED sink driver

Subtotal ≈ £2.70 ex VAT (≈ £3.24 inc VAT; DigiKey UK ships free above its threshold, otherwise ≈ £12).

2. Other purchases (not Farnell)

Item	Qty	Supplier	Code / search	GBP each	Note
LiPo pouch 3.7 V 1200 mAh, protected, JST-PH 2.0 lead (64 x 35.5 x 5.3 mm)	2	Pimoroni	BAT0004	8.25	Alternative: The Pi Hut 105188 (62 x 35 x 5 mm). Meter polarity: + must land on J2 pin 1 (silk +).
Gooseneck 120 mm, 10 mm OD, M10x1 male ends, hollow (6.8 mm bore)	2	Amazon UK (PATIKIL 4.7in M10 gooseneck)	search: PATIKIL gooseneck M10 120mm	8.00	Backup: PATIKIL 100 mm M10/M8 (8 mm OD; bore unstated - VERIFY). Check bore >= 4.5 mm.
Silicone tube 10 mm ID x 12 mm OD, 150 mm (sleeve for the gooseneck: the plastic-coated finish)	2	Amazon UK / eBay (food-grade silicone tube)	10x12mm silicone tube 1 m	6.00	Or 12 mm heat-shrink 2:1 in black/white.
M10x1 brass hex nut, ~14 mm AF, 3 mm thick (moon base: one in the cup's socket, one jam nut under the floor)	4	Lamp Spares UK / any lamp-parts shop	M10x1 lamp nut	0.50	Often supplied with the gooseneck. Measure AF/thickness (VERIFY in cad/parts/_lib/moonlib.py). The moon-end thread must be >= 9 mm long (VERIFY M1).
M10 flat washer, <= 20 mm OD, ~1 mm thick (under the cup floor, above the jam nut)	2	Lamp Spares UK / Accu.co.uk	M10 washer (lamp) or M10 form A	0.20	Brass or steel; the PETG floor is clamped between it and the socket nut.
Qwiic / STEMMA QT 4-pin JST-SH cable >= 200 mm (cut one end; bare end solders to the LED board)	2	RS Components	288-4665 (SparkFun KIT-15081 Qwiic cable kit, 50-500 mm)	11.00	Kit has several lengths: use the 200 or 500 mm one. Farnell alternative: SHR-04V-S-B housings + pre-crimped leads.
M2 x 6 mm self-tapping pan-head screws (PCB to front shell)	4	Amazon/Accu.co.uk	M2x6 self tapping pan head A2	0.10	Accu.co.uk sells singles.
M2 x 12 mm self-tapping pan-head screws (clip + back plate to front shell)	4	Amazon/Accu.co.uk	M2x12 self tapping pan head A2	0.10	
Filament: white PLA (moon) ~12 g, PETG (base, pack, clip, caps) ~40 g	2	Bambu Lab (PLA Basic Jade White, PETG Basic)		1.50	Per lamp, filament cost.
Cyanoacrylate (super glue, gel) for the gooseneck ends and the LED-pad strain relief	1	any		3.00	Gooseneck threads only. NOT anaerobic thread-locker: it does not cure against plastic and can craze PETG (round1 m-7).
Neutral-cure silicone (or E6000): strain relief over the LED-board wire joints and the cable at J3	1	any		5.00	Not acetoxy (vinegar-smell) silicone: it corrodes copper.

Item	Qty	Supplier	Code / search	GBP each	Note
PTFE tube 2 mm ID x 3 mm OD (or silicone sleeve), 200 mm: protects the cable inside the gooseneck	2	Amazon UK / 3D-printer spares	PTFE tube 2x3mm	0.50	Plus 6 mm heat-shrink over each cut gooseneck end (round1 M-7).
Kapton (polyimide) tape 10-20 mm: over the pouch under the PCB, under U1/D3/Q1-Q3, and a disc over the LED board's LP5012 (faces the gooseneck's metal end)	1	any		3.00	One roll covers many builds (round1 B-2).
USB-C charger >= 1 A at 5 V (if you don't have one)	1	any		0.00	The lamp can draw ~0.5-0.6 A while charging with the light on (round1 M4).

Other subtotal ≈ £84.70.

3. Tools (once)

Item	Supplier	Code	GBP	Note
UPDI programmer: Microchip MPLAB SNAP (PG164100)	Farnell	2915518	15.00	Or a 3.3 V USB-UART (FTDI TTL-232R-3V3, Farnell 1329311) + 470 R for SerialUPDI.
Stencils (PCBWay/JLCPCB, frameless): main top 0.12 mm, LED board top 0.12 mm + bottom 0.10 mm	with the PCB order		12.00	Skip the LED bottom stencil if the fab assembles that face.

4. PCBs (handoff 01; route comparison in `bom/assembly_routes.md`)

Main board: 5 bare boards with a stencil, hand assembly (route 1). LED board: 5 boards with the **bottom face (U1 LP5012 WQFN, U2, R1, C1–C4) assembled by JLCPCB (Economic, ≈ £10–25 incl. extended-part fees)**, then hand-fit the top; or bare with both stencils if you already do QFNs. Boards with stencils: PCBWay ≈ £40–60 incl. shipping, JLCPCB ≈ £15–35. Upload `fab/main-board/v0.1/main-board-v0.1-gerbers.zip` and `fab/led-board/v0.2/led-board-v0.2-gerbers.zip`, and copy the specs from each `FAB_NOTES.md`.

5. Total for this build (main board by hand, LED board bottom face by JLCPCB)

Farnell ≈ £47 + DigiKey ≈ £15 + other ≈ £85 + programmer ≈ £15 (if needed) + PCBs ≈ £15–60 + LED bottom-face assembly ≈ £10–25 → **≈ £160–247 for 2 lamps.** (If the fab sources the LP5012 from LCSC, drop the DigiKey line.)